Dong 2020 prostate cancer — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

All 4 malignant-bearing samples (≥915 cells each) intermix in one connected manifold — single kNN component, 0 detached islands; iLISI median rises 1.00→1.28/4 under integration (one patient is ~49% of cells, capping the scalar). Judged on the embedding across a θ×n_pcs sweep, not a scalar.

Gate 2 — module geography → GO (non-branching)

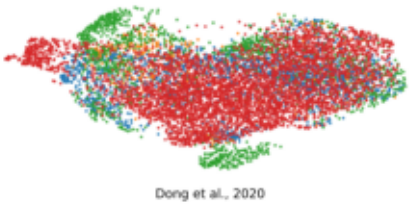
Single Gast/pSom → Crest axis. Gast & pSom spatially OVERLAP and co-vary (r=+0.47) — not a pSom-arm vs Gast-arm split; both oppose Crest (Gast–Crest −0.38, pSom–Crest −0.40). Along pseudotime Gast/pSom fall (pSom ρ=−0.41) and Crest rises (ρ=+0.39) — an inverse terminal gradient, not a hub-and-spoke bifurcation.

Gate 3 — cross-repression → NA

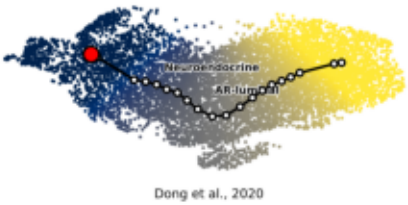
Single monotone axis — no bifurcation and no Gast/pSom-overlap "Mixed" hub with divergent arms, so there is no branch pair to test. (Spatially-defined hub Gast↔pSom r≈−0.04, i.e. no cross-repression; the apparent −0.38 under sum-thresholding is a collider/selection artifact, not competition.)

STEP FIGURES

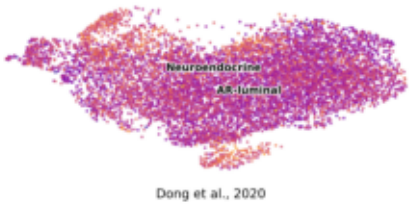
Manifold — samples



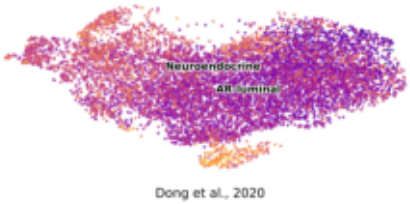
Pseudotime



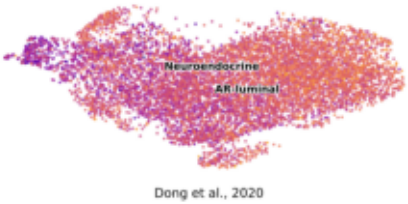
Gastrulation



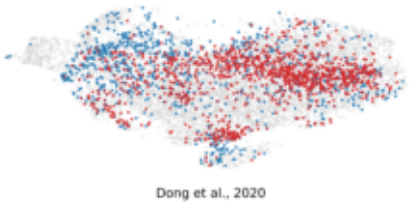
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.

Prostate cancer (Dong et al., 2020), 10x — 9,698 malignant cells across 4 samples (≥10x floor keeps all; no clean count gap). 8,281 enter the trajectory after a cross-patient cycling compartment (1,417 cells) is scoped out; the developmental manifold is a single connected component.

Verdict.

Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. The malignant manifold is a single monotone gastrulation/somitogenesis → terminal-neural-crest gradient, a strict linear subset of Puram's hub-and-spoke: Gast & pSom co-vary and overlap rather than diverging into competing arms, so there is no Mixed hub and no bifurcation to test. Raw module means Gast +0.035 / pSom +0.100 / Crest −0.041. An orthogonal AR-luminal ↔ neuroendocrine lineage axis (decoupled from clinical histology) is continuous and multi-patient — not a second trajectory. Curve is fold-free (self-approach 1.74; independent-diffusion continuity −0.79). EMT-TFs: SOX9 (early/root) → TWIST1 (terminus), detection-gated, VIM excluded.

Parameter choices

Min cells/sample	≥100 (no count gap; all 4 samples 915–4765)	**Integration**	Harmony θ=10; PCA 50; spread=1.3
Scope	cycling lobe excluded (1,417) → 8,281 dev	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	Gast/pSom-high tip (Crest-low origin)	**Embedding**	n_pcs=20; overdispersed=1,670
Geometry	Linear (non-branching); 2 tips / 0 forks	**EMT-TFs**	SOX9 (early/root) → TWIST1 (terminus); hub r NA